

MET CS688

WEB ANALYTICS AND MINING

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AUTOGEN

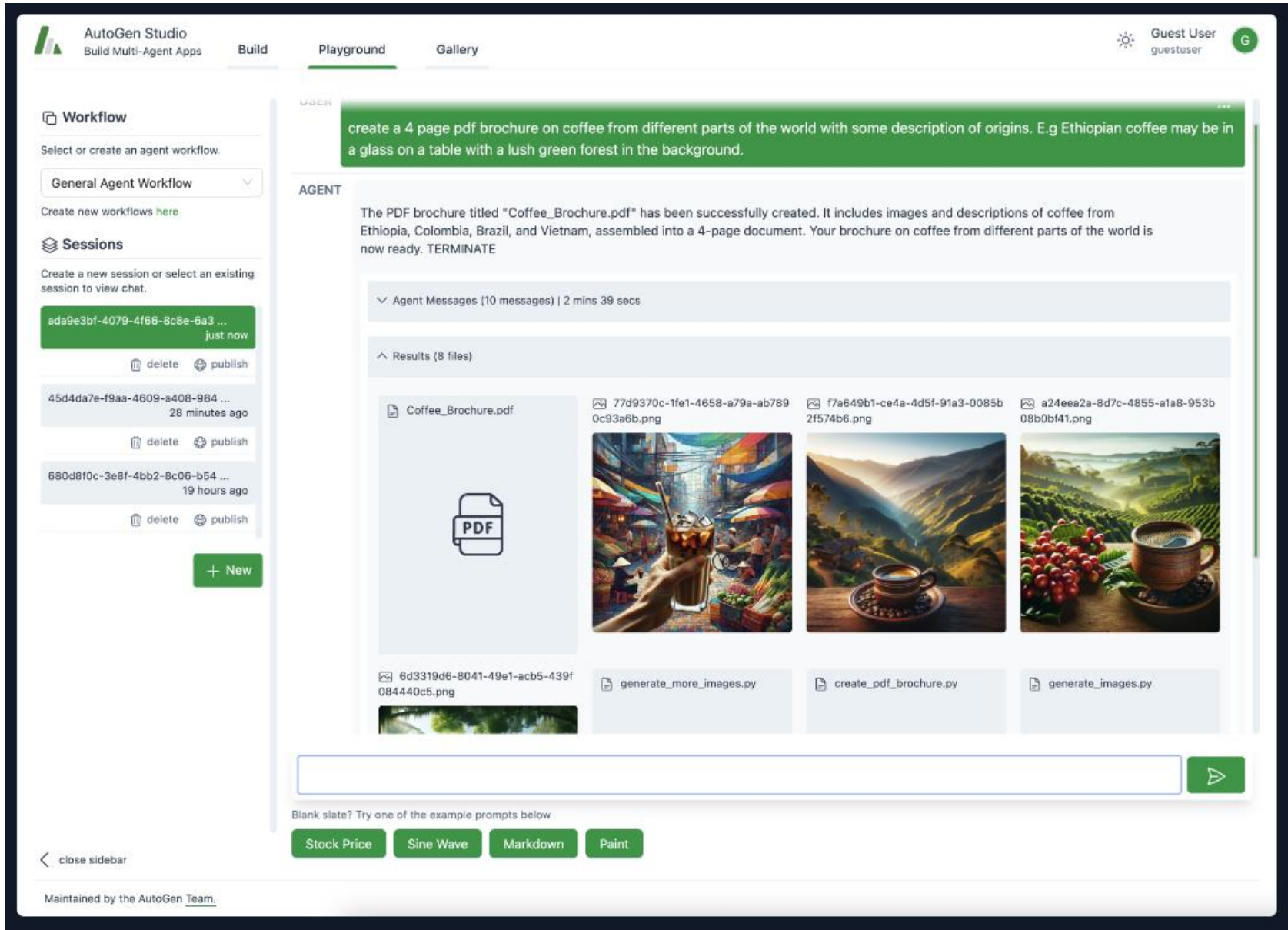
What is AutoGen?

AutoGen Studio UI is a wide-ranging tool for AI development.

It transforms complexity into a user-friendly experience.

Intuitive creating of AI agents.

AutoGen Rag



What is AutoGen?

The Multi-AI Agent era started when OpenAI released Swarm.

AutoGen is an open-source Python-based framework from Microsoft Research that allows users to define, configure, and compose AI agents for multi-agent applications.

AutoGen's features include:

- Multi-agent collaboration: Agents can work together to solve tasks
- LLM and tool support: Agents can use large language models (LLMs) and tools
- Human-in-the-loop workflows: Agents can easily integrate human feedback and involvement
- Consensus-building: Agents can incorporate diverse viewpoints

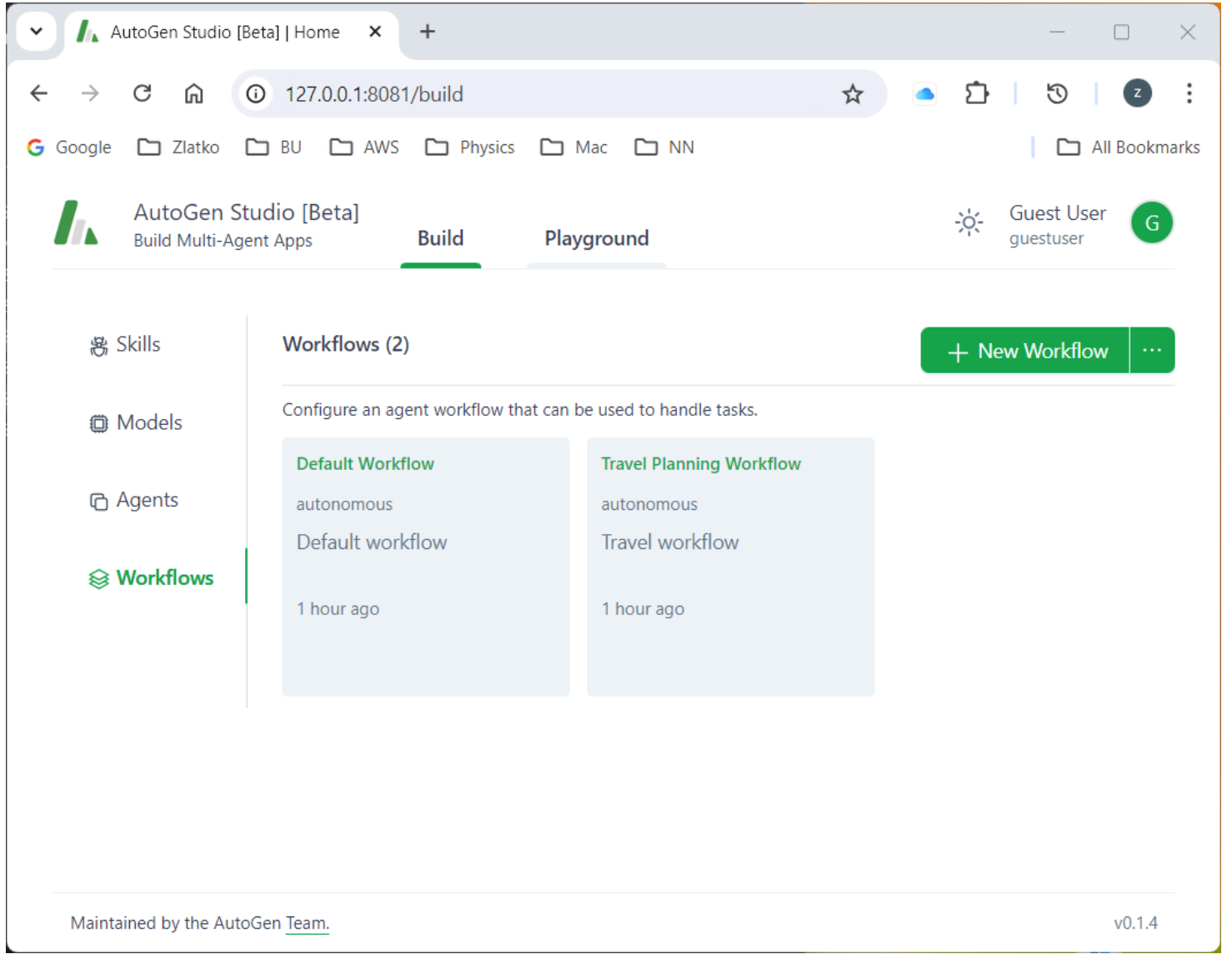
Applications you can build with AutoGen:

- You can use AutoGen to build a wide range of applications from various domains and complexities, such as
    - code generation,
    - data analysis,
    - summarization, etc.

Unlike AutoGen, AutoGPT does not have the functionality to have multiple agents collaborating to produce better results.

1) Installing AutoGen Studio

- Create a new project & virtual environment
- Install package: autogenstudio
- Launching AutoGen Studio:
  - To start the AutoGen Studio user interface in the terminal type:
    autogenstudio ui
  - In a browser open http://localhost:8081/.
  - This is the URL where AutoGen Studio is running.



Creating Skills - Exploring AutoGen Studio Features

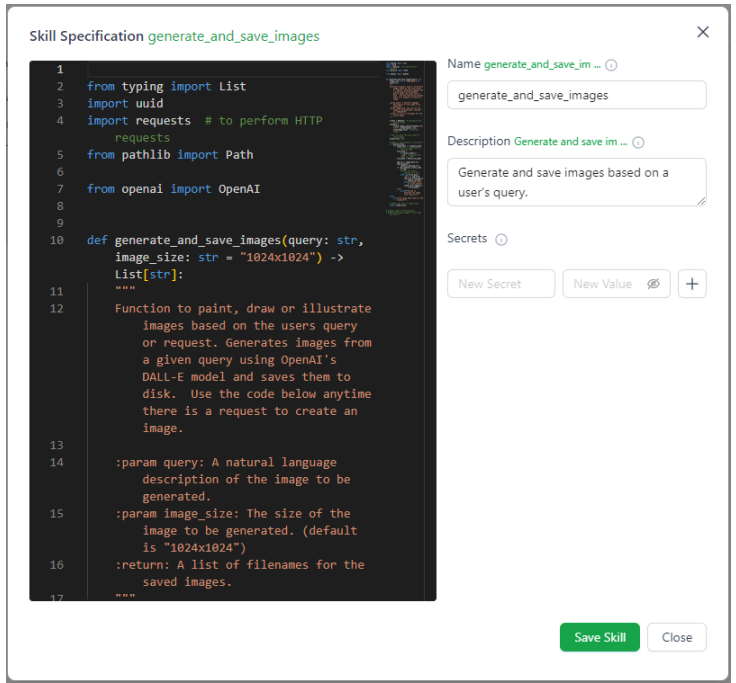
The **Build** section, is a starting point in crafting an effective AI solution. Here you can experiment, create, and fine-tune the elements that make up your AI solution.
- Gives you the opportunity to create new skills (**Skills** are python functions that agents can use to solve tasks.).

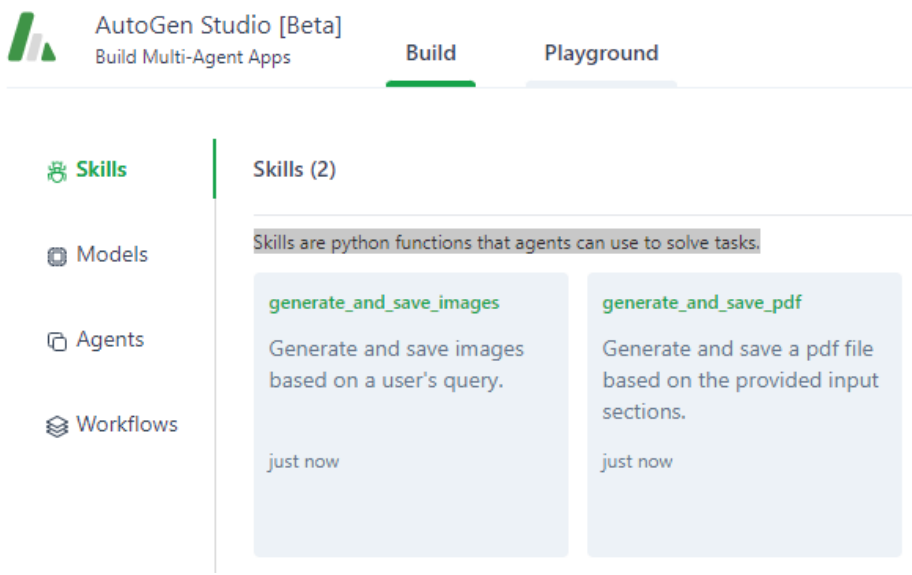
Existing ones include:

- Generate and save images based on a user's query.
- Generate and save a pdf file based on the provided input sections.

Skills (programming a specific action for your AI agent.)

- Creating (or Uploading)





Creating Agents - Exploring AutoGen Studio Features

The “**Agents**” tab is where you create your AI agents.

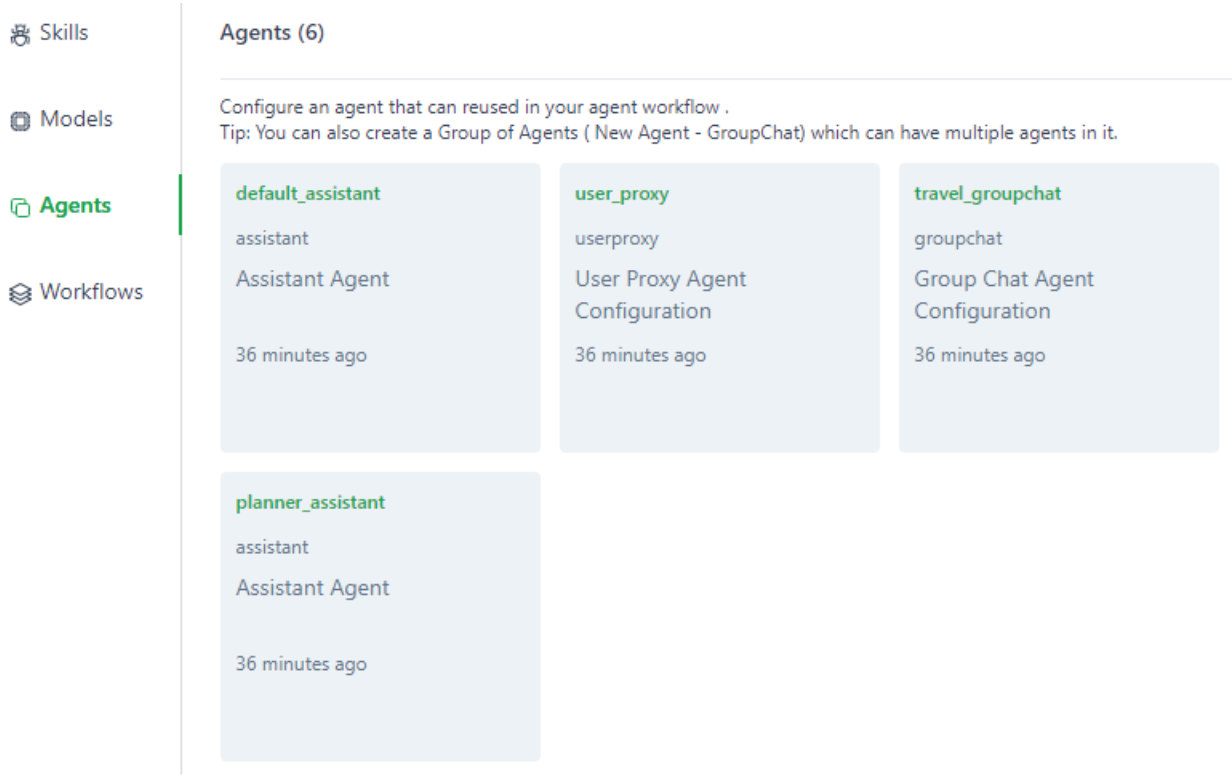
- Here, you can create as many agents as your project requires, each with unique characteristics and skills.

For instance, you might create a sample agent and customize it with a specific model like “gp4 turbo.”

- This customization includes defining the agent’s behavior, responses, and capabilities.

Adding **skills** to your agents is a straightforward process.

- You can choose from skills you’ve created or use predefined skills available in AutoGen Studio.

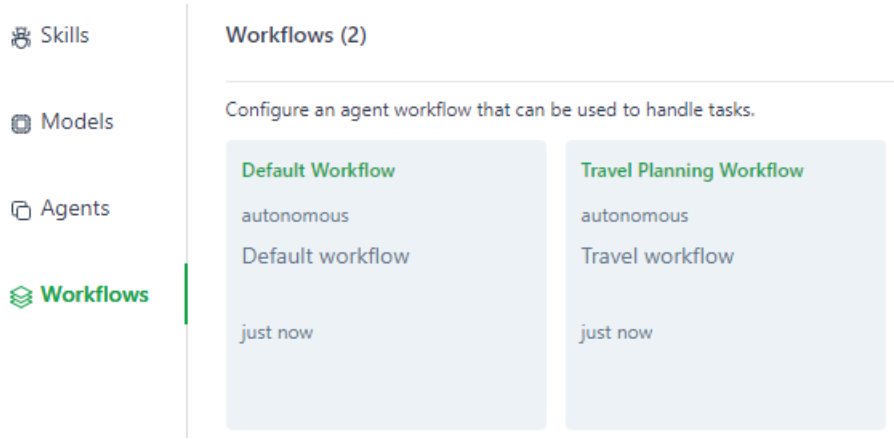


Creating Workflows - Exploring AutoGen Studio Features

Workflows are the essence of interaction within your AI ecosystem. They define how different agents collaborate to perform tasks.

Setting up a workflow involves specifying the interaction patterns between agents.

- For example, you might establish a workflow where two agents work together, exchanging information and decisions to complete a task.
- AutoGen Studio allows interactions between agents.



Playground - Exploring AutoGen Studio Features

The **Playground** section is where the action happens.

- It's the space where you can see the workflows in action and agent's collaboration comes to life.
- Interact with them and observe the results of your configurations from the Build section.

Interacting with Agent Workflows:

- Dynamic process where you can see how your agents collaborate, use their skills, and adapt to different scenarios (i.e. agents collaborating to generate images).

Testing and experimenting:

- You can run example tasks to see how your agents and workflows perform.
- See how the agents interact, how they utilize their skills, and how effectively they accomplish the set objectives.

Session: A **session** in the Playground refers to the entire period of interaction with the agent workflow.

- Within a session, there's also the **Chat** View.
    - It enables conversation with your AI agent, guiding it, and receiving feedback in real-time.

Utilizing the AutoGen Studio API

Declarative Workflow Specification:

- The API allows for agent workflows to be declaratively specified in JSON format.
- This feature is particularly useful for defining workflows programmatically, providing the flexibility to customize and automate processes to a greater extent.

Workflow Management:

- Through the API, you can load and run JSON-defined workflows.
- The process involves creating a workflow configuration from your agent specifications and then utilizing the AutoGenWorkFlowManager to manage and execute these workflows.

2) LLM Providers

AutoGen Studio supports a wide range (open source or commercial) of large language model (LLM) providers.

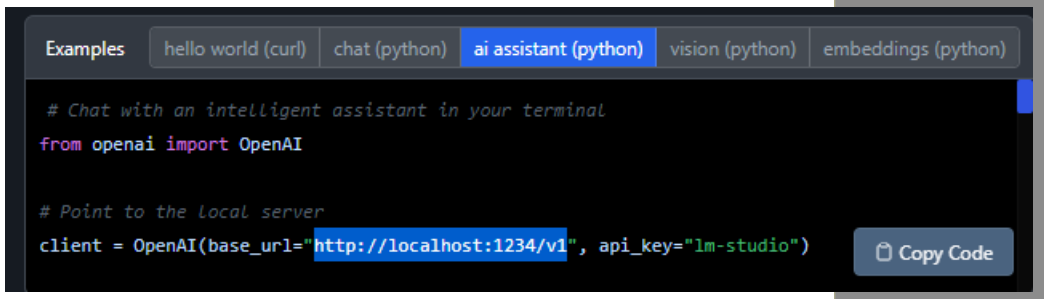
- AutoGen Studio platform seamlessly integrates with models from Ollama and LM Studio.
- Supports integration with OpenAI's API, enabling access to some of the most cutting-edge language models available.

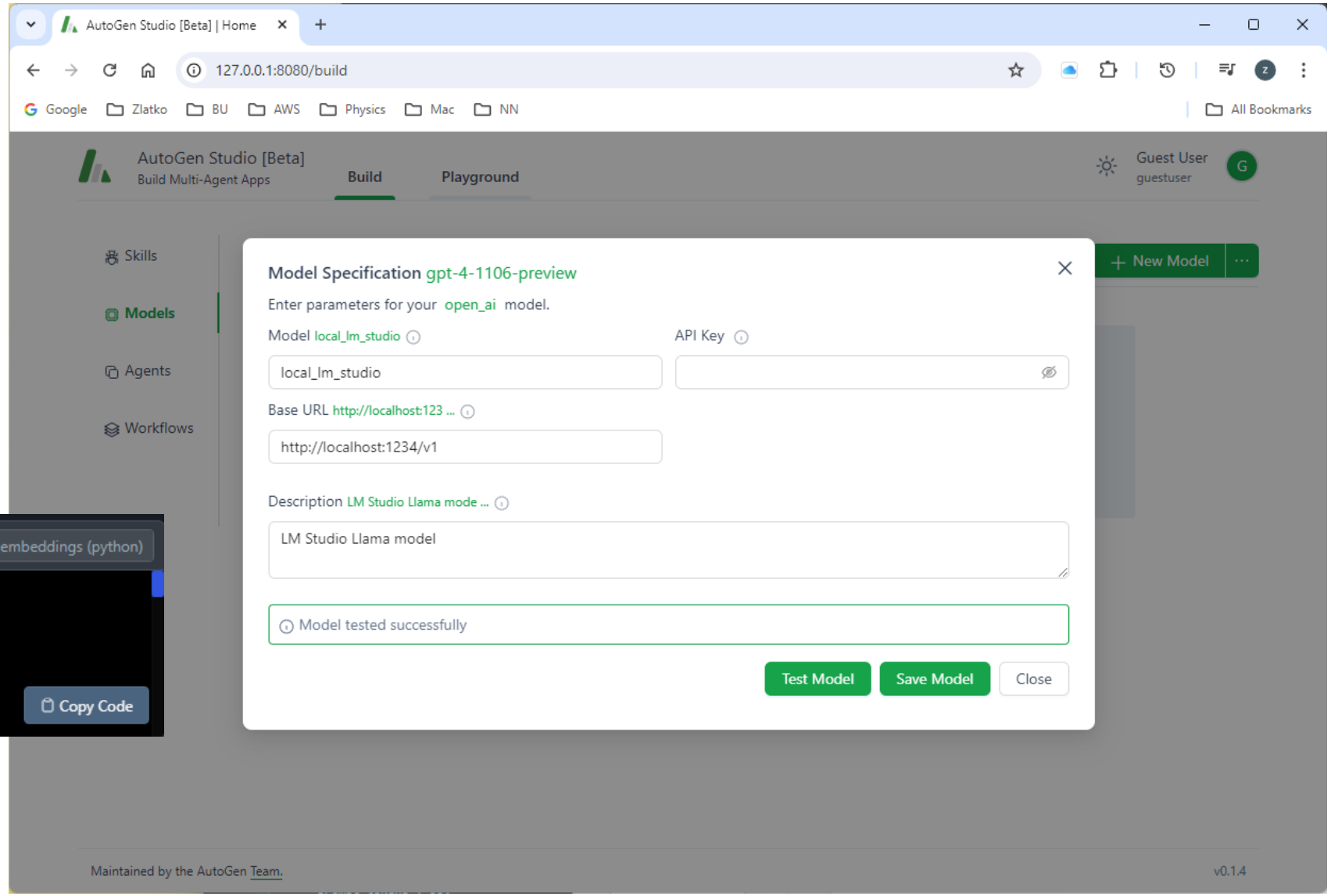
Integration with LM Studio

- LM Studio is an easy-to-use desktop app for experimenting with local and open-source Large Language Models (LLMs).

Add New Model from LM Studio

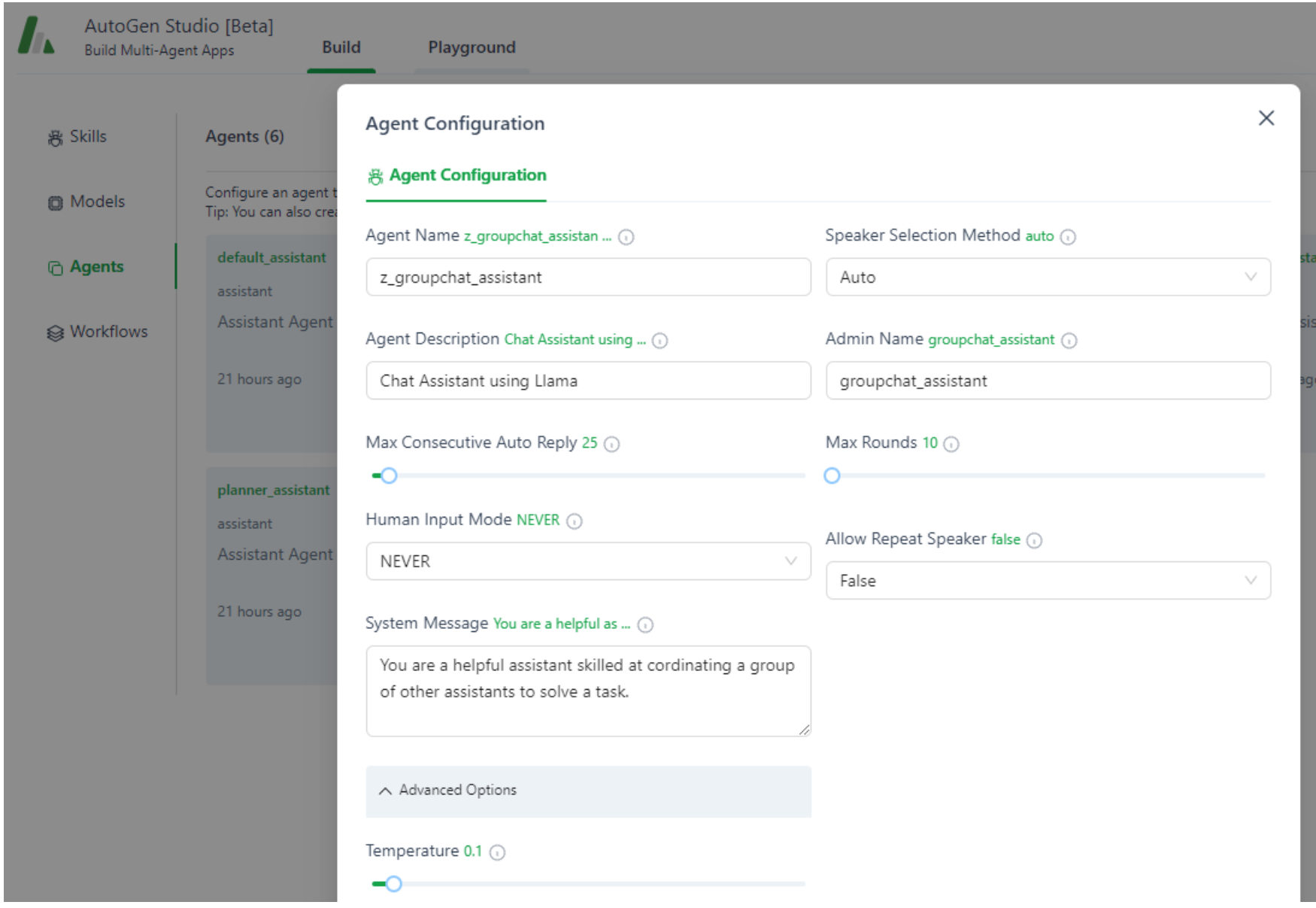
- Select **Models**, then **+New Model** and populate configuration.
- Provide Name
- API Key is not needed for local LLM
- Provide Base URL by coping it from LM Studio (i.e. http://localhost:1234/v1).





Creating New Agents

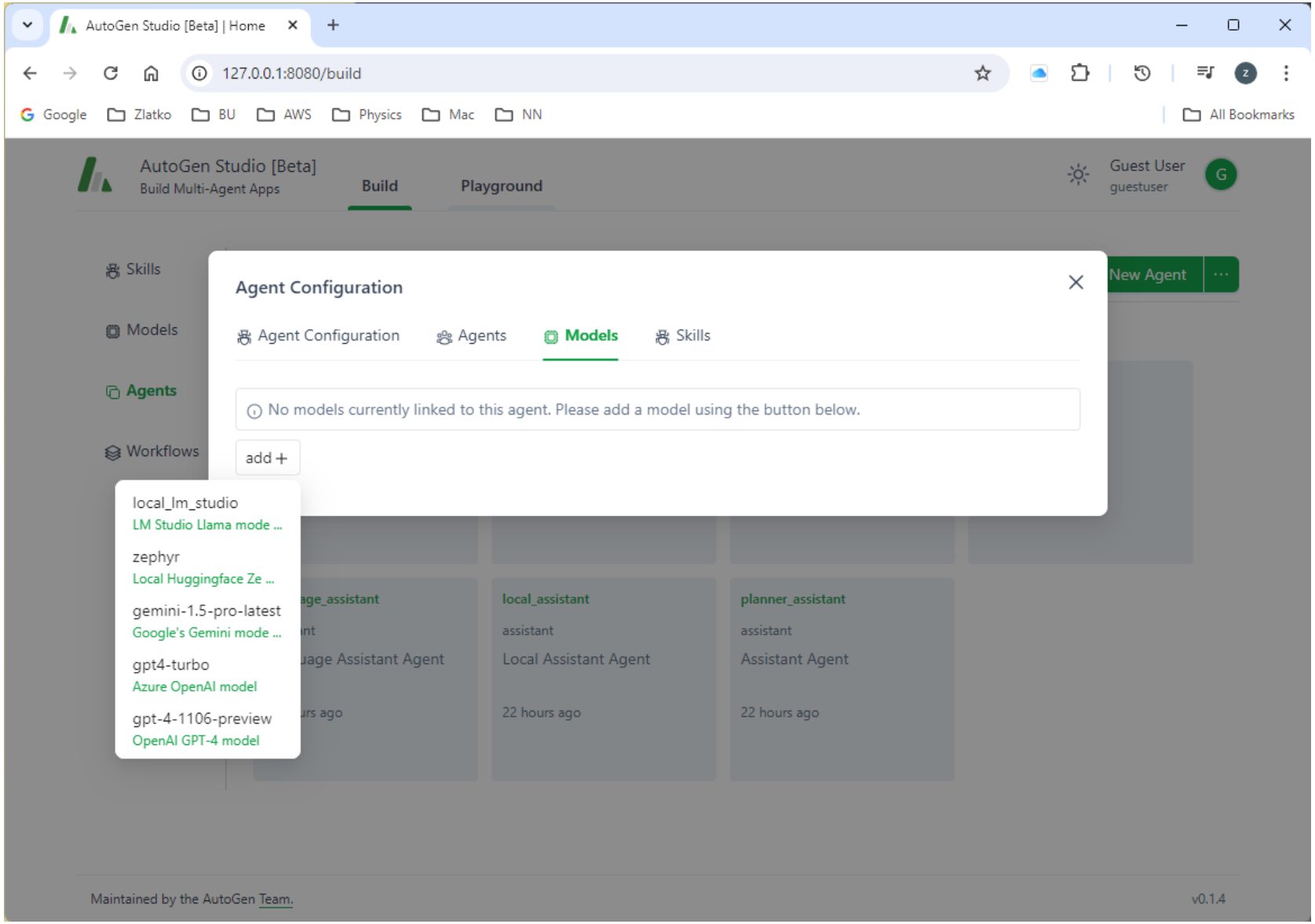
- Select **Agents**, then **+New Agent** and populate configuration.
- You can also create a Group of Agents (New Agent - GroupChat) which can have multiple agents in it.



Add Model to New Agent

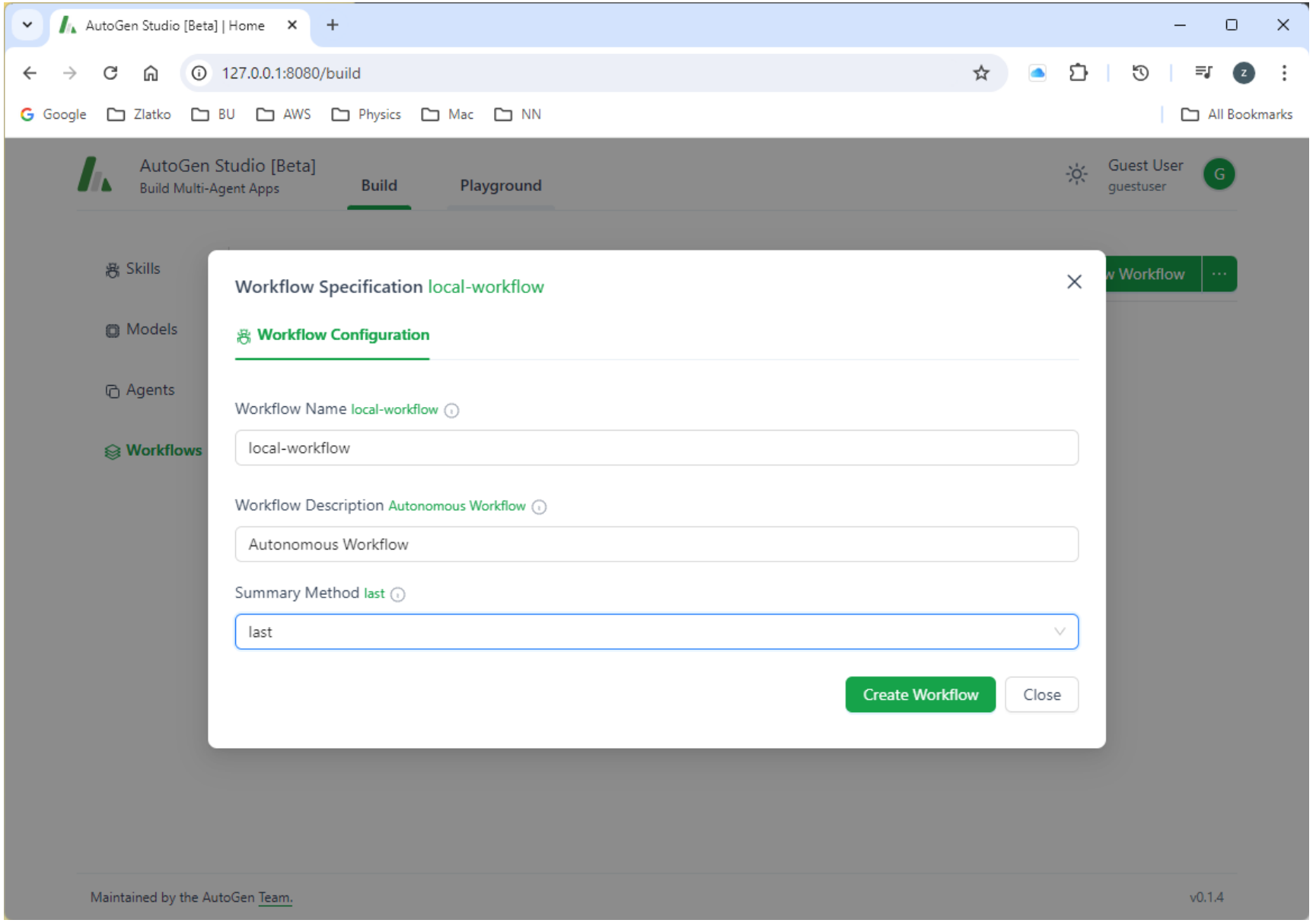
Add **Model** to the **New Agent**

- One that you previously created
- Existing one (i.e. GPT)



Add New Workflow

Add **New Workflow**



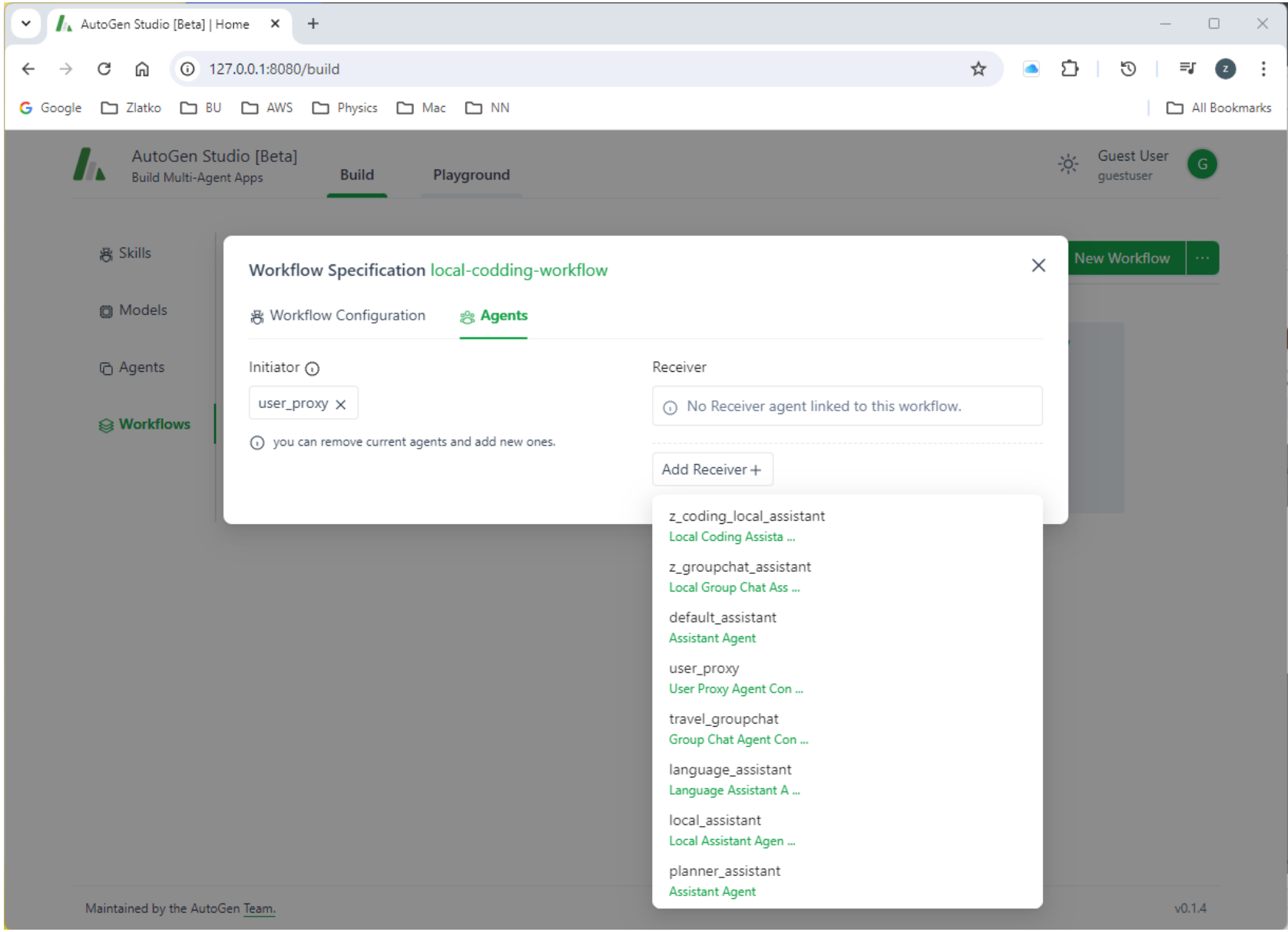
Specify Agents for the New Workflow

Specify **Agents** for the **New Workflow**

Keep **Initiator** as user_proxy

Choose **Receiver** as

- Either the one you created
- Or choose from existing ones.



Add Existing Skills and Start New Session

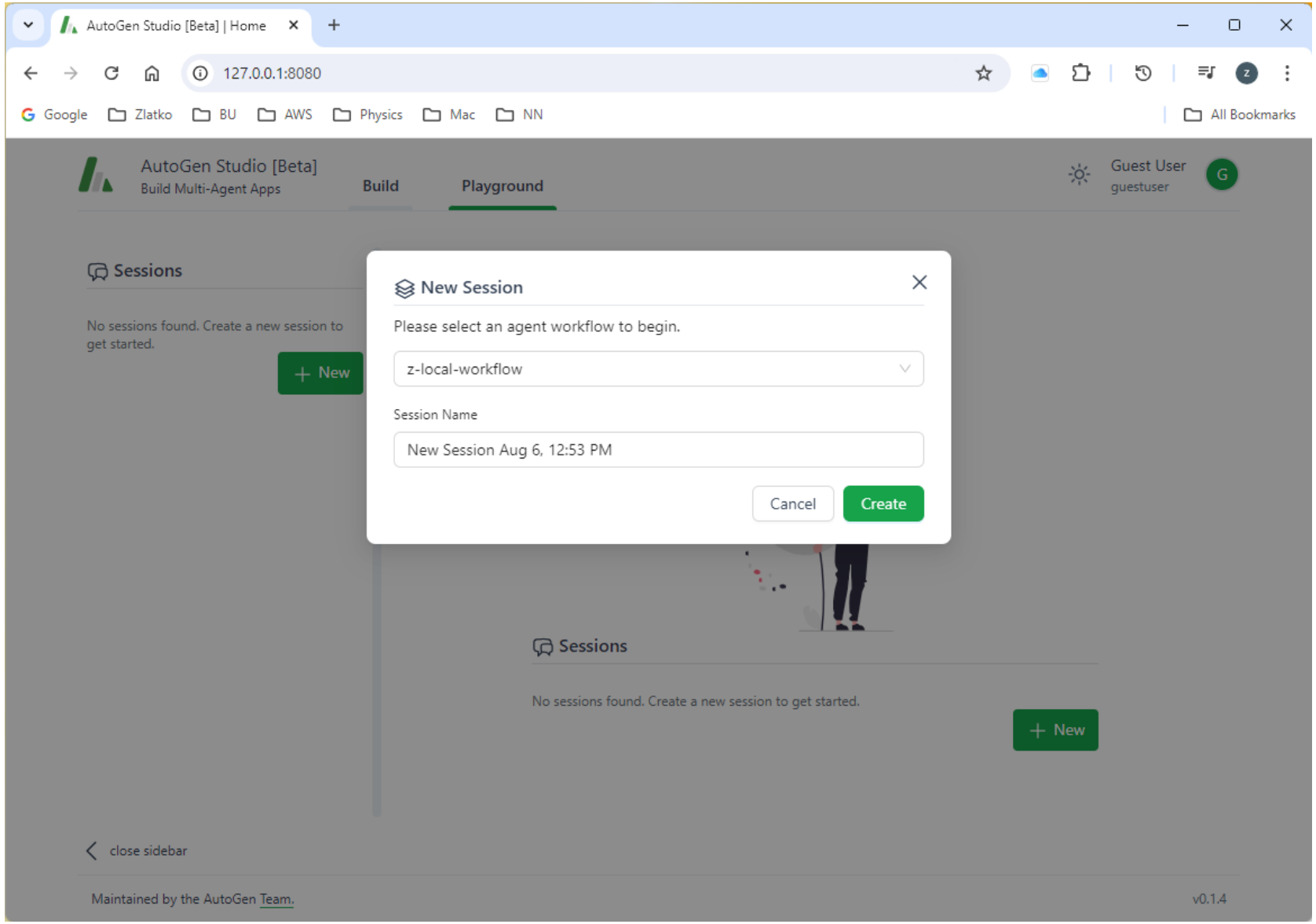
From **Agents** tab, open newly created Agent and from the existing **Skills** add one to it. The default Skills are:

- Generate and save images
- Generate and save PDF

Note: To generate an image, you need to use existing Models such as GPT.

To generate images with local Model, you need to use stable diffusion model (typically used for image generation) and define it as a new Skill.

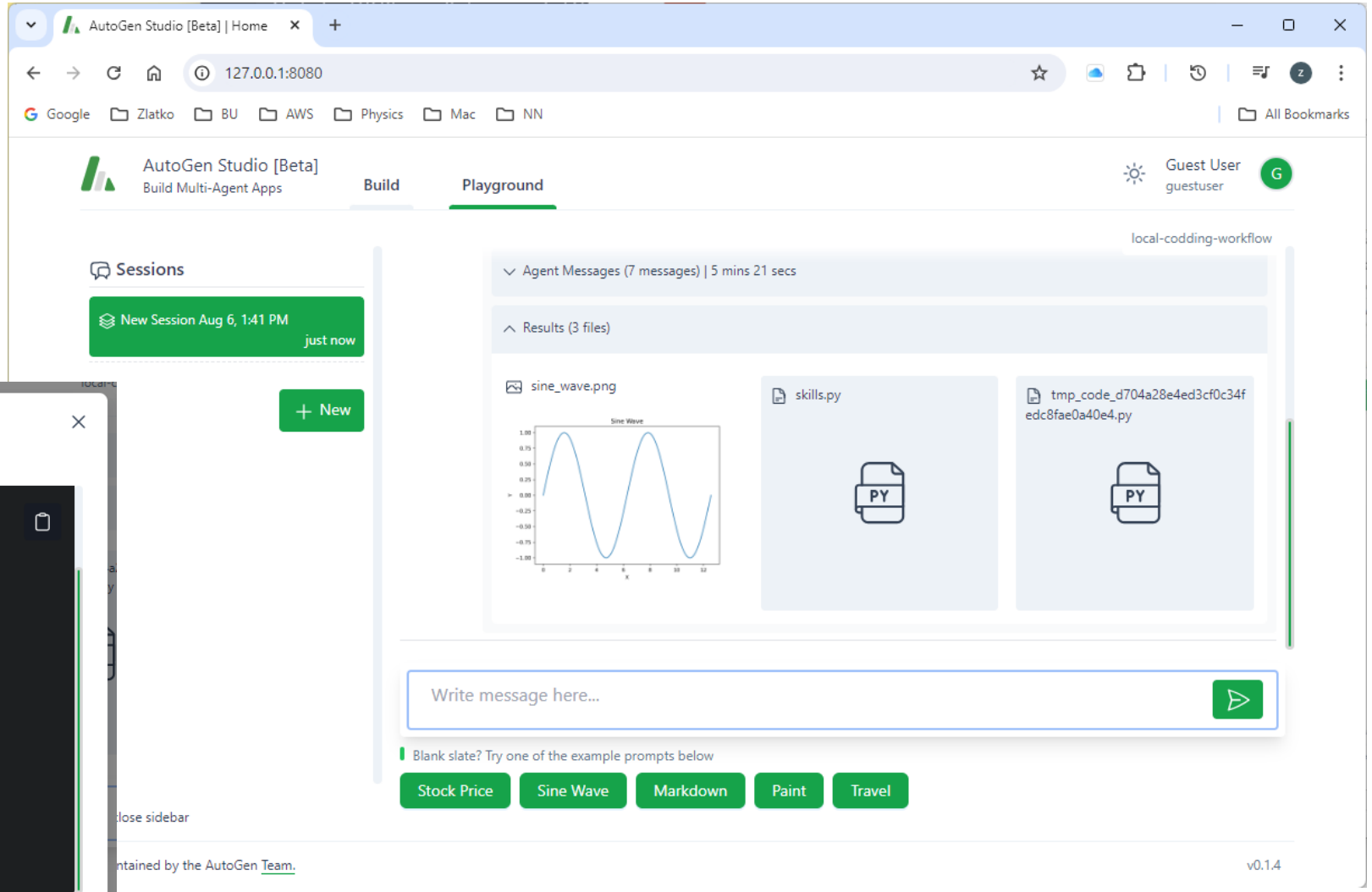
Start a new session in **Playground** to test the newly created workflow.



Test New Workflow

From the **New Session** try one of the default prompts such as:

Write a python script to plot a sine wave and save it to disc as a png file sine_wave.png





Source: https://microsoft.github.io/autogen/docs/autogen-studio/usage/

Export AutoGen Studio Workflows to a Python App

- An exported workflow can be easily integrated into any Python application using the WorkflowManager class with just two lines of code.

```
from autogenstudio import WorkflowManager
# load workflow from exported json workflow file.
workflow_manager = WorkflowManager(workflow="path/to/your/workflow_.json")

# run the workflow on a task
task_query = "What is the height of the Eiffel Tower?. Dont write code, just respond to the
question."
workflow_manager.run(message=task_query)
```

- Deploying AutoGen Studio Workflows as APIs
  - The workflow can be launched as an API endpoint from the command line using the autogenstudio command line tool.

```
autogenstudio serve --workflow=workflow.json --port=5000
```

- Similarly, the workflow launch command above can be wrapped into a Dockerfile that can be deployed on cloud services like Azure Container Apps or Azure Web Apps.

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What is Magentic-One?

Microsoft Magentic-One is a New Multi-AI Agent Framework, which is an extension to AutoGen.

- Magentic-One is a Multi-agent system that:
  - Handles various web and file-based tasks across multiple fields.
  - It's built to help with tasks similar to those people face in daily work or personal settings.
  - An o**pen-source** version of Magentic-One is available on Microsoft's AutoGen framework.
- AutoGen vs Magentic-One - ease of use
  - **Foundation:** Magentic-One is built on the AutoGen framework, which provides the underlying architecture for multi-agent communication and cooperation. AutoGen is a standalone framework
  - **Purpose:** Magentic-One is a generalist multi-agent system designed to handle complex, multi-step tasks using specialized agents, whereas AutoGen serves as a broader framework for creating various types of AI agents without a specific focus on task execution.
  - **Agent Structure:** Magentic-One features a structured approach with an Orchestrator agent managing four specialized agents (WebSurfer, FileSurfer, Coder, ComputerTerminal) to collaboratively complete tasks. AutoGen does not prescribe such a specific architecture for its agents.
  - **Evaluation Tools:** Magentic-One includes AutoGenBench, a dedicated evaluation tool to assess agent performance on benchmarks like GAIA and AssistantBench. While AutoGen allows for the creation of agents, it does not include built-in performance evaluation tools.
  - **Flexibility and Adaptability:** Magentic-One's modular architecture supports the easy addition or removal of agents without affecting overall performance, promoting adaptability. AutoGen provides the foundational capabilities but may require more manual adjustments for similar adaptability.
  - **Model Agnosticism:** Both systems are model-agnostic; however, Magentic-One is currently optimized for specific models like GPT-4o while allowing integration with various language models based on task needs. AutoGen is designed to support multiple models generally but does not focus on specific implementations.

What is Magentic-One?

Microsoft Magentic-One has these 5 default agents:

- **Orchestrator Agent:** The central component responsible for high-level planning and task management. It directs other agents, tracks progress, and can re-plan if progress stalls.

- **Specialized Agents:** Four additional agents support the Orchestrator:
  - ***WebSurfer:*** *Manages a web browser for navigation and interaction with webpages.*
  - ***FileSurfer:*** *Handles local file management and navigation.*
  - ***Coder:*** *Specializes in writing and analyzing code.*
  - ***ComputerTerminal:*** *Provides console access for executing programs and installing libraries.*









Link: https://www.microsoft.com/en-us/research/articles/magentic-one-a-generalist-multi-agent-system-for-solving-complex-tasks/

What is Magentic-One?

How it works?

This diagram illustrates the workflow of the Orchestrator system, focusing on the **Task Ledger** and **Progress Ledger** to manage task progress:

1. **Task Ledger**: The Orchestrator starts by creating or updating this ledger with key information like known facts, assumptions, and task plans.
2. **Progress Ledger**: At each step, the Orchestrator updates this ledger to check task completion, monitor progress, and note if any agents need reassigning.
3. **Stall Detection**: If the task stalls for more than two cycles, the Orchestrator revises the Task Ledger and plan.
4. **Agent Coordination**: The Orchestrator assigns subtasks to agents (Coder, ComputerTerminal, WebSurfer, and FileSurfer), who act based on the instructions provided.
5. **Completion Check**: Once progress is satisfactory, the system checks for completion and reports the final result.

